

Data Knowledge

BRD Where clause



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What is the WHERE Clause in SQL?

The **WHERE clause** is used in SQL to **filter records** from a table based on **specific conditions**. It allows you to **retrieve only the rows that meet the criteria** you define.

Operator	Definition / Use Case	Example
=	Checks if a column value is equal to a specified value.	salary = 60000
>	Checks if a column value is greater than a specified value.	salary > 60000
<	Checks if a column value is less than a specified value.	salary < 50000
>=	Checks if a column value is greater than or equal to a specified value.	salary >= 70000
<=	Checks if a column value is less than or equal to a specified value.	salary <= 65000
!= or <>	Checks if a column value is not equal to a specified value.	department != 'IT'
BETWEEN	Selects values within a range , inclusive of boundaries.	salary BETWEEN 50000 AND 70000
NOT BETWEEN	Selects values outside a specified range.	salary NOT BETWEEN 50000 AND 70000
IN	Matches any value from a list of specified values.	department IN ('IT', 'HR')
NOT IN	Excludes rows where the value is in the list.	position NOT IN ('Intern', 'Support')
LIKE	Used for pattern matching with wildcard characters: % (any sequence), _ (1 char).	first_name LIKE 'A%'
NOT LIKE	Finds values not matching the specified pattern.	first_name NOT LIKE 'A%'

Operator	Definition / Use Case	Example
IS NULL	Checks if a column contains a NULL (missing/empty) value.	position IS NULL
IS NOT NULL	Checks if a column does not contain NULL .	salary IS NOT NULL
AND	Combines multiple conditions where all must be true .	salary > 50000 AND department = 'IT'
OR	Combines conditions where at least one must be true .	department = 'HR' OR salary > 70000
NOT	Reverses the result of the condition that follows it.	NOT department = 'HR' or NOT (salary > 50000)

SELECT column1, column2, ...
FROM table_name
WHERE condition;

Data

-- Create the employee table

```
CREATE TABLE employee (  
  emp_id INT, first_name  
  VARCHAR(50), last_name  
  VARCHAR(50), gender  
  VARCHAR(10), department  
  VARCHAR(50), position  
  VARCHAR(50), salary  
  DECIMAL(10,2), join_date  
  DATE
```

);

-- Insert 20 sample records into employee

INSERT INTO employee VALUES

(1, 'John', 'Doe', 'Male', 'IT', 'Developer', 60000.00, '2022-01-10'),
(2, 'Jane', 'Smith', 'Female', 'HR', 'Manager', 75000.00, '2021-03-15'),
(3, 'Bob', 'Brown', 'Male', 'Finance', 'Analyst', 58000.00, '2020-07-20'),
(4, 'Alice', 'Johnson', 'Female', 'Marketing', 'Executive', 62000.00, '2022-05-18'),
(5, 'David', 'Lee', 'Male', 'IT', 'Tester', 50000.00, '2023-02-22'),
(6, 'Emma', 'Wilson', 'Female', 'HR', 'Recruiter', 48000.00, '2022-11-30'),
(7, 'Michael', 'Clark', 'Male', 'Finance', 'Manager', 82000.00, '2019-09-10'),
(8, 'Olivia', 'Lopez', 'Female', 'IT', 'Developer', 61000.00, '2023-04-04'),
(9, 'Daniel', 'Lewis', 'Male', 'Marketing', 'Designer', 55000.00, '2021-08-14'),
(10, 'Sophia', 'Hall', 'Female', 'IT', 'Support', 47000.00, '2020-12-25'),
(11, 'Liam', 'Young', 'Male', 'HR', 'Coordinator', 49000.00, '2023-01-05'),
(12, 'Isabella', 'King', 'Female', 'Finance', 'Accountant', 56000.00, '2021-10-01'),
(13, 'Noah', 'Scott', 'Male', 'IT', 'Architect', 95000.00, '2018-06-17'),
(14, 'Mia', 'Green', 'Female', 'Marketing', 'Strategist', 68000.00, '2022-03-27'),
(15, 'Ethan', 'Adams', 'Male', 'Finance', 'Intern', 35000.00, '2024-01-10'),
(16, 'Ava', 'Baker', 'Female', 'IT', 'Intern', 36000.00, '2024-03-15'),
(17, 'James', 'Nelson', 'Male', 'HR', 'Trainer', 54000.00, '2020-04-20'),
(18, 'Charlotte', 'Carter', 'Female', 'Marketing', 'Executive', 60000.00, '2019-11-11'),
(19, 'Benjamin', 'Mitchell', 'Male', 'Finance', 'Consultant', 88000.00, '2021-05-07'), (20,
'Amelia', 'Perez', 'Female', 'IT', 'Developer', 63000.00, '2023-07-01');

Employee Data Query Module

Business Context:

The **HR Analytics Team** of a mid-size IT services company requires a dynamic query system to **filter and retrieve employee data** for internal reports, appraisals, audits, hiring trends, salary analysis, and department-level reviews.

The system should allow business analysts and managers to run SQL-based queries on the employee table using various filtering conditions such as department, salary range, date of joining, gender, and job roles.

Questions:

1. Select all employees from the IT department.
2. Get employees whose salary is more than 60,000.
3. List employees who joined before 2022.
4. Find employees with position = 'Developer'.
5. Show employees whose gender is 'Male'.
6. Get employees from the HR department.
7. Select employees with salary between 50,000 and 70,000.
8. List employees not in the Finance department.
9. Retrieve employees who joined in 2023.
10. Show employees whose first name starts with 'J'.
11. Find employees whose last name ends with 'n'.
12. Get all employees where salary is less than 40,000.
13. Select employees where salary is not equal to 60,000.
14. List employees where position is either 'Manager' or 'Developer'.
15. Get all employees who joined after '2022-01-01' and have a salary > 60,000.
16. Find employees from departments 'IT', 'HR', or 'Finance' using IN.
17. Select employees not from departments 'Marketing' and 'HR'.
18. Get employees whose first name contains the letter 'a'.
19. Show employees whose position is not 'Intern'.
20. List employees who joined between '2020-01-01' and '2022-12-31'.
21. Find employees whose salary is not between 50,000 and 70,000.
22. Select employees who joined in 2023 and are from the IT department.
23. Retrieve employees whose salary is >= 60,000 and position = 'Developer'.
24. Find employees with last names that have exactly 5 characters.
25. Show employees whose first name starts with 'A' or ends with 'e'.
26. List employees whose department is 'Marketing' and salary < 60,000.
27. Get employees where salary is > 50,000 OR department = 'HR'.

28. Find employees whose position is 'Executive' and joined before '2021-01-01'.
29. Select employees who are not from 'Finance' and salary is above 55,000.
30. Get all employees whose position contains the word 'Manager' using LIKE.
31. Find employees whose name starts with the same letter as their last name.
32. Get employees who joined in the last 2 years and are not Interns.
33. Select employees whose department is 'IT' or 'Finance' but not 'HR'.
34. List employees who joined on "27-12-2024"
35. Retrieve employees whose salary is between 50,000 and 70,000 **excluding 60,000**.
36. Find employees where first name does not contain vowels.
37. Get employees where either: department is IT and salary > 60,000 or department is HR and joined after 2022
38. Select employees with position = 'Manager' or (salary > 80,000 and not in IT).
39. Find employees whose join date is **not** between '2021-01-01' and '2023-12-31'.
40. Get employees where position is in ('Executive', 'Strategist') and salary < 65,000 OR joined before '2020-01-01'.

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